

Construction and Status of the Canonical Spin-1 Deuteron GTMD/TMD Model

A light-front, flavor-resolved, spin-resolved, quark–gluon boundary before complete rank-aware evolution

DeuteronWigner project

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Scope statement. This is the authoritative scientific and implementation note for the leading-twist forward boundary at $Q = 5$ GeV and $x_N \in \{0.02, 0.05, 0.10, 0.20, 0.40\}$. The model contains all declared spin-1 quark, antiquark, and gluon projections, retains proton and neutron parents, and separates exact constraints, fitted inputs, model assumptions, and unresolved physics. Complete common-scheme multi- Q evolution is the next phase and is not claimed here.

Abstract

We describe the construction, physical content, validation, and present status of a canonical light-front model for the complete declared leading-twist spin-1 deuteron transverse-momentum-dependent (TMD) system. The organizing object is a retained quark or gluon parton–target helicity correlator obtained from a nucleon correlator and a spin-dependent light-front deuteron spectral density. Named TMDs are projections of this parent, rather than independent curves. The boundary is explicitly resolved in $u, d, \bar{u}, \bar{d}$, proton and neutron source, target polarization U, L, T, LL, LT, TT , orbital interference, nuclear mechanism, and, for naive-T-odd gluons, the antisymmetric f^{abc} and symmetric d^{abc} gauge-link/color classes.

The numerical construction combines CT18 and MSHT20QED unpolarized inputs, BDSSV24 helicity replicas, JAMDiFF plus lattice-informed transversity, BPV20 Sivers replicas, a published-interval treatment of the Yang g_{1T} fit, physically constrained Boer–Mulders, pretzelosity, worm-gear, tensor, and gluon sectors, six realistic deuteron wave functions, binding and Fermi motion, off-shell response, shadowing and antishadowing, a sourced $NN\pi$ component, and controlled non-nucleonic interfaces. We give the field-theoretic definitions, polarization and transverse-rank algebra, nuclear convolution, composition rules, uncertainty semantics, and validation conditions. We also document why two earlier implementations—a real one-body boundary and a smooth reduced-amplitude closure—were insufficient. The accepted result is a physically organized pre-evolution boundary, not a global extraction and not a claim of equal evidential precision for all TMDs.

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1 Scientific objective and status language

The objective is not a minimally runnable atlas. It is a model in which all realistically supported contributions to the spin-1 partonic density can coexist without losing their origin or being counted twice. A model is called *canonical* here when

1. the dynamical state is a retained quark–target or gluon–target correlator;
2. every named TMD is obtained by a declared projection of that correlator;
3. proton, neutron, flavor, helicity, wave-function component, gauge-link, color, and nuclear-mechanism identity remain available until the physical observable is assembled;
4. a machine-readable composition graph distinguishes baselines, additive mechanisms, mutually exclusive alternatives, and ensemble members;
5. exact constraints, fit information, lattice information, and model assumptions remain different provenance classes.

The word *complete* must always carry a scope. In this note it means: the complete declared leading-twist forward quark and gluon basis at the fixed boundary scale $Q = 5$ GeV, with smooth functions and named bands over the five production x_N nodes. It does not mean a first-principles solution of QCD, a simultaneous global fit, complete nonzero-skewness GTMD phenomenology, or complete TMD evolution.

1.1 Evidence vocabulary

We use the following classification throughout:

Exact	A consequence of the chosen operator definition, representation, discrete symmetry, kinematics, or algebraic composition rule.
Fit/phenomenology	A central function or ensemble tied to a published PDF/TMD extraction or directly calibrated observable.
Lattice-informed	A fit or model component constrained by lattice matrix elements or an explicit lattice prior.
Model	A physically motivated but underdetermined component with a replaceable parameterization and a named sensitivity ensemble.
Temporary	A controlled implementation boundary retained only until a specified replacement, with a test that distinguishes the two.

Passing a structural acceptance gate does not upgrade a model prediction to a fit.

2 From the original proposal to the present model

2.1 The GTMD-first proposal

The original technical draft, *A GTMD-First Light-Front Wigner and SCET Framework for Spin-1 Deuteron TMDs*, proposed the correct parent chain, following the GTMD/Wigner logic of Refs. [9, 10]:

$$\Psi_D^{\text{LF}} \longrightarrow \rho_{\Lambda'\Lambda;\lambda'\lambda}^{N/D} \longrightarrow W_{a/N}^{[\Gamma]} \longrightarrow W_{a/D}^{[\Gamma]} \longrightarrow \Phi_{a/D}^{[\Gamma]} \longrightarrow F_{a/D}. \quad (1)$$

The light-front deuteron wave function Ψ_D^{LF} determines the active-nucleon spin density $\rho^{N/D}$; this density convolves with a nucleon partonic GTMD $W_{a/N}$; the forward correlator $\Phi_{a/D}$ is projected onto named functions $F_{a/D}$.

The parent structure is important because the reductions must commute:

$$F_{a/D}(x, \mathbf{k}_T; \eta) = W_{a/D}(x, \mathbf{k}_T, \Delta_T = \mathbf{0}; \eta), \quad \text{TMD limit,} \quad (2)$$

$$H_{a/D}(x, \Delta_T) = \int d^2 \mathbf{k}_T W_{a/D}(x, \mathbf{k}_T, \Delta_T), \quad \text{zero-skewness GPD limit,} \quad (3)$$

$$f_{a/D}(x) = \int d^2 \mathbf{k}_T F_{a/D}(x, \mathbf{k}_T), \quad \text{rank-zero PDF limit,} \quad (4)$$

$$\mathcal{W}_{a/D}(x, \mathbf{k}_T, \mathbf{b}_\Delta) = \int \frac{d^2 \Delta_T}{(2\pi)^2} e^{-i\mathbf{b}_\Delta \cdot \Delta_T} W_{a/D}(x, \mathbf{k}_T, \Delta_T), \quad \text{partonic Wigner transform.} \quad (5)$$

The nuclear Wigner/spectral density and the partonic Wigner distribution are distinct: the first describes nucleons in the deuteron; the second is the transverse Fourier transform of a QCD partonic correlator.

2.2 First refocusing: exact zeros were boundary zeros

The first numerical parent used a real, helicity-independent, rank-zero nucleon GTMD. It correctly tested Fourier signs, recoil, hermiticity, marginals, and the tensor PDF entering b_1 . It could not represent the complete spin structure. If all amplitudes are real,

$$\text{Im} \left[\psi_{L'_z}^* \mathcal{U}_\eta \psi_{L_z} \right] = 0 \quad \implies \quad F_{\text{naive-T-odd}} = 0, \quad (6)$$

and missing off-diagonal helicity or OAM amplitudes similarly force worm-gear, pretzelosity, and tensor projections to vanish. These were controlled one-body limits, not predictions of exact physical zeros.

2.3 Second refocusing: smooth closure was not parent dynamics

A later reduced-amplitude construction populated every TMD name with a smooth curve and conservative rank factors. Its map was schematically

$$\{f_1, g_1, h_1, f_{1LL}\}_{\text{anchors}} \longrightarrow \{\mathcal{A}_U, \mathcal{A}_L, \mathcal{A}_T, \mathcal{A}_{LL}, \mathcal{A}_{LT}, \mathcal{A}_{TT}\} \longrightarrow \{F_A\}_{\text{all names}}. \quad (7)$$

This was useful for registry and plotting tests but bypassed equation (1). Shared shapes and coefficients produced flavor-independent appearances, while proton and neutron sources were summed too early. Passing smoothness and individual positivity ceilings did not establish a physical model. Those outputs are now explicitly superseded.

2.4 Third refocusing: isoscalar equality belongs at final assembly

In the exact charge-symmetric one-body limit, an inclusive $I = 0$ deuteron satisfies

$$u_D = u_{p/D} + u_{n/D} = u_{p/D} + d_{p/D} = d_D, \quad \bar{u}_D = \bar{d}_D. \quad (8)$$

This does not imply $u_p = d_p$, nor does it justify erasing constituent identity. The production state therefore retains

$$\{\text{proton-in-deuteron, neutron-in-deuteron, nucleon sum,} \\ \text{proton-minus-neutron, nuclear correction, spin-1 total}\}.$$

The spin-1 total is a closure projection, not the stored model state.

2.5 Fourth refocusing: basis parity was not evidence parity

After the correlator basis was populated, a final audit required every one of the 36 declared quark-plus-gluon projections to identify a central source, constituent resolution, uncertainty or sensitivity ensemble, CSB status, shared-parent projection, channel-appropriate nuclear dressing, and a physical benchmark or controlled limit. All 36 rows now pass this evidence-discipline gate. Many tensor and gluon rows remain predictions, not direct extractions.

3 Kinematics and spin-1 representation

We use lightlike vectors $n^\mu, \bar{n}^\mu$ with $n^2 = \bar{n}^2 = 0$ and $n \cdot \bar{n} = 1$:

$$v^\mu = v^+ \bar{n}^\mu + v^- n^\mu + v_T^\mu, \quad v^+ = v \cdot n, \quad v^- = v \cdot \bar{n}. \quad (9)$$

At zero skewness the average deuteron momentum is P , the transverse transfer is Δ_T , and the active parton has average transverse momentum $\mathbf{k}_T$. The staple direction is denoted $\eta = +$ (future) or $\eta = -$ (past). Nuclear convolutions use the nucleon-scaled fraction x_N , with $x_D = x_N/2$ in the project convention.

3.1 Target density and irreducible basis

The target Hilbert space is $\mathcal{H}_D = \text{span}\{|+1\rangle, |0\rangle, |-1\rangle\}$. Every Hermitian target matrix has the irreducible decomposition

$$\text{Herm}(3) = \underbrace{U}_1 \oplus \underbrace{L}_1 \oplus \underbrace{(T_x, T_y)}_2 \oplus \underbrace{LL}_1 \oplus \underbrace{(LT_x, LT_y)}_2 \oplus \underbrace{(TT_x, TT_y)}_2. \quad (10)$$

The dimensions sum to nine. With trace-orthogonal basis matrices B_A , a target matrix M is reconstructed as

$$M = \sum_A c_A B_A, \quad c_A = \sum_B (G^{-1})_{AB} \text{Tr}(B_B^\dagger M), \quad G_{AB} = \text{Tr}(B_A^\dagger B_B). \quad (11)$$

This is the concrete representation-theoretic machinery used in the code. It prevents incompatible target channels from being mixed and exposes unidentifiable combinations through the Gram rank.

For helicity-resolved scalar quantities F_Λ , useful combinations are

$$F_U = \frac{1}{3}(F_+ + F_0 + F_-), \quad F_L = \frac{1}{2}(F_+ - F_-), \quad \delta_T F = F_0 - \frac{1}{2}(F_+ + F_-). \quad (12)$$

The internal LL basis has the opposite sign to the physical Trento S_{LL} convention; explicit adapters are therefore required for f_{1LL} and $h_{1LL}^\perp$.

3.2 Transverse rank

The transverse plane carries $SO(2)$. A rank- r coefficient $F^{(r)}$ multiplies an irreducible harmonic $k_T^{i_1 \dots i_r}$, and its physical modulation scales as

$$\mathcal{M}_r(x, \mathbf{k}_T) \sim \left(\frac{k_T}{M_D} \right)^r F^{(r)}(x, k_T^2). \quad (13)$$

A positive-rank coefficient can be finite at $k_T = 0$, while the physical modulation must vanish. Rank determines the Fourier–Bessel kernel:

$$\tilde{F}^{(r)}(x, b_T) = 2\pi \int_0^\infty k_T dk_T J_r(b_T k_T) \mathcal{N}_r(k_T) F^{(r)}(x, k_T), \quad (14)$$

$$F^{(r)}(x, k_T) = \int_0^\infty \frac{b_T db_T}{2\pi} J_r(b_T k_T) \tilde{\mathcal{N}}_r(b_T) \tilde{F}^{(r)}(x, b_T). \quad (15)$$

The registry stores r and the mass convention so that a scalar J_0 transform cannot be applied to rank-one or rank-two functions.

4 Operator correlators and gauge links

4.1 Quark and antiquark correlators

For flavor q , the unsubtracted zero-skewness quark GTMD correlator is

$$W_{\Lambda'\Lambda}^{q[\Gamma]}(x, \mathbf{k}_T, \mathbf{\Delta}_T; \eta) = \frac{1}{2} \int \frac{dz^- d^2 \mathbf{z}_T}{(2\pi)^3} e^{ixP^+ z^- - i\mathbf{k}_T \cdot \mathbf{z}_T} \langle P', \Lambda' | \bar{q}(-z/2) \Gamma \mathcal{U}_\eta[-z/2, z/2] q(z/2) | P, \Lambda \rangle \Big|_{z^+=0}. \quad (16)$$

The leading-twist projections are $\Gamma = \gamma^+$, $\gamma^+ \gamma_5$, and $i\sigma^{i+} \gamma_5$ [4, 5]. The physical TMD is UV-renormalized and soft-subtracted. Antiquarks are represented by a separate positive- x sector rather than by silently identifying them with quarks.

At fixed $(x, \mathbf{k}_T)$, the retained quark object is a 3×3 target matrix for each Dirac projection, equivalently a 6×6 joint target–quark helicity density when the quark helicity indices are made explicit.

4.2 Gluon correlator

The gluon correlator and its gauge-link labels follow the general spin ≤ 1 classification of Ref. [7]:

$$\Gamma_{\Lambda'\Lambda}^{ij}(x, \mathbf{k}_T, \mathbf{\Delta}_T; \eta_1, \eta_2) = \frac{1}{xP^+} \int \frac{dz^- d^2 \mathbf{z}_T}{(2\pi)^3} e^{ixP^+ z^- - i\mathbf{k}_T \cdot \mathbf{z}_T} \times \langle P', \Lambda' | \text{Tr}_c \left[F^{+i}(-z/2) \mathcal{U}_{\eta_1} F^{+j}(z/2) \mathcal{U}_{\eta_2} \right] | P, \Lambda \rangle \Big|_{z^+=0}. \quad (17)$$

The transverse 2×2 gluon space decomposes into trace (unpolarized), antisymmetric (circular polarization), and symmetric-traceless (linear polarization) pieces. Together with the target space, the stored forward parent is a $3 \times 3 \times 2 \times 2$ joint matrix.

4.3 Discrete symmetries and process dependence

Hermiticity requires

$$W_{\Lambda'\Lambda}(x, \mathbf{k}_T, \mathbf{\Delta}_T; \eta) = W_{\Lambda\Lambda'}^*(x, \mathbf{k}_T, -\mathbf{\Delta}_T; \eta). \quad (18)$$

Parity is implemented as a tested light-front helicity reflection. Time reversal maps future to past gauge links:

$$F_{\text{even}}^{[+]} = F_{\text{even}}^{[-]}, \quad F_{\text{odd}}^{[+]} = -F_{\text{odd}}^{[-]}. \quad (19)$$

For gluons, the two universal naive-T-odd color contractions

$$\mathcal{C}_f^{abc} = if^{abc}, \quad \mathcal{C}_d^{abc} = d^{abc} \quad (20)$$

remain independent. A process observable has the form

$$F_{\text{process}}^g = C_f^{\text{process}} F^{g(f)} + C_d^{\text{process}} F^{g(d)}. \quad (21)$$

The model refuses an implicit universal f/d mixture or a mixed-link sign assigned by quark analogy.

5 Complete declared leading-twist basis

The current forward basis follows Refs. [6, 5, 7]. It has 18 quark functions for each of $u, d, \bar{u}, \bar{d}$, and 18 identifiable gluon projections. The older exploratory 19-gluon count split a TT pair that the implemented forward correlator identifies only through a combination; it is not the canonical registry.

Table 1: Declared quark and antiquark basis for each flavor. The rank is the rank of the transverse tensor multiplying the named scalar coefficient.

Target	Named functions	Physical content
U	$f_1, h_1^\perp$	unpolarized density; Boer–Mulders
L	$g_1, h_{1L}^\perp$	helicity; longitudinal worm gear
T	$f_{1T}^\perp, g_{1T}, h_1, h_{1T}^\perp$	Sivers, transverse worm gear, transversity, pretzelosity
LL	$f_{1LL}, h_{1LL}^\perp$	longitudinal tensor density and linear/chiral-odd partner
LT	$f_{1LT}, g_{1LT}, h_{1LT}, h_{1LT}^\perp$	mixed tensor polarization
TT	$f_{1TT}, g_{1TT}, h_{1TT}, h_{1TT}^\perp$	transverse tensor polarization

Table 2: Declared gluon basis. The f-type and d-type gauge-link/color sectors are additional labels on T-odd entries, not additional TMD names.

Target	Named functions	Physical content
U	$f_1^g, g_1^g, h_1^{\perp g}$	unpolarized, helicity, linear polarization
L	$h_{1L}^{\perp g}$	longitudinal spin–linear-polarization interference
T	$f_{1T}^{\perp g}, g_{1T}^g, h_1^g, h_{1T}^{\perp g}$	gluon Sivers, worm gear, and transverse spin/linear structures
LL	$f_{1LL}^g, h_{1LL}^{\perp g}$	longitudinal tensor gluons
LT	$f_{1LT}^g, g_{1LT}^g, h_{1LT}^g, h_{1LT}^{\perp g}$	mixed tensor gluons
TT	$f_{1TT}^g - h_{1TT}^{\perp g}, g_{1TT}^g, h_{1TT}^g, h_{1TT}^{\perp g}$	identifiable transverse-tensor combinations

6 Nucleon-level quark dynamics

6.1 Unpolarized and helicity sectors

The central collinear f_1^q uses CT18NNLO [15] with distinct $u, d, \bar{u}, \bar{d}$ proton functions. The 29 CT18 Hessian pairs are propagated through the resolved nuclear response on all five production x_N nodes. The neutron is not obtained by assigning the same function to both flavors. Exact charge symmetry supplies the baseline rotation

$$u_n = d_p, \quad d_n = u_p, \quad \bar{u}_n = \bar{d}_p, \quad \bar{d}_n = \bar{u}_p, \quad (22)$$

and MSHT20QED [21] supplies a numerical unpolarized CSB ratio

$$\delta_q^n(x, Q) = \frac{q_{\text{MSHT20QED}}^n(x, Q)}{q_{\text{MSHT20QED}}^{p, \text{isospin partner}}(x, Q)} - 1 \quad (23)$$

with 38 paired Hessian directions. This numerical correction is not copied to polarized or T-odd channels.

Quark g_1 uses BDSSV24-NLO [16]. All 600 replicas are propagated through the AV18 response and through the production x grid. The wave-function spin depolarization emerges from the helicity convolution; it is not implemented only as a universal scalar.

6.2 Transversity and chiral-odd worm gear

The h_1 input is the JAMDiFF w LQCD ensemble [17]: member zero plus 968 physical replicas. The combination is fit and lattice informed. Because the source fit and the project f_1, g_1 use different underlying PDF sets, each member is projected into the actual TMD-level Soffer interval

$$|h_1^q(x, k_T)| \leq \frac{1}{2} [f_1^q(x, k_T) + g_1^q(x, k_T)] \quad (24)$$

with a small numerical interior margin. Raw source members remain recoverable, so this compatibility projection is visible and replaceable.

The $h_{1L}^\perp$ boundary is correlated with the same replica identity through a Wandzura–Wilczek-type functional. Schematically,

$$h_{1L}^{\perp(1)q}(x) \simeq -x^2 \int_x^1 \frac{dy}{y^2} h_1^q(y) + \Delta_{\text{genuine}}^q(x). \quad (25)$$

The first term is an organized approximation; the genuine quark–gluon term is an independent model sensitivity, not absorbed into the fit band.

6.3 Sivers and Boer–Mulders

The quark Sivers function uses the BPV20 N3LO arTeMiDe release [14] with 500 replicas. Member identity is propagated through all six nuclear wave functions. The source fit’s light-sea equality is retained as a documented fit assumption rather than generalized into a project-wide flavor identity. Staple reversal follows equation (19).

Boer–Mulders $h_1^\perp$ is represented by flavor-dependent sign/hierarchy scenarios informed by the fitted Sivers rescattering strength. The relation

$$h_1^{\perp q}(x, k_T) = C_{\text{BM}}^q(x, k_T) f_{1T}^{\perp q}(x, k_T) \quad (26)$$

is explicitly **Model**: C_{BM}^q has alternative members and is not a claim of an operator identity or a joint fit.

6.4 Worm gear g_{1T} and pretzelosity

The central g_{1T} parameters follow Yang et al. (2024) [18]. Because no public covariance or replica ensemble is available, the implementation propagates all 16 corners of the four published asymmetric parameter intervals. The resulting hull is not labeled a confidence interval. A shared-Fock/OAM sea member tests the source fit's zero-sea boundary.

Pretzelosity is not generated by rescaling transversity; its unusual small- b_T status is discussed in Ref. [19]. Its nonperturbative first transverse moment is parameterized inside

$$h_{1T}^{\perp(1)q}(x) = \int d^2\mathbf{k}_T \frac{k_T^2}{2M_N^2} h_{1T}^{\perp q}(x, k_T^2), \quad \left| h_{1T}^{\perp(1)q} \right| \leq \frac{f_1^q - g_1^q}{2}. \quad (27)$$

The central perturbative small- b_T boundary is zero, while signed hadronic OAM members use a conservative fraction of equation (27). These members are model sensitivities.

6.5 Tensor-polarized quarks

The ten LL/LT/TT quark functions are projections of the common deuteron spin-density parent. Their sources are S-D and OAM interference, Wilson channels, spin recoupling, and operator-valued nuclear response. In particular, g_{1LT} and g_{1TT} require absorptive axial-tensor interference and are implemented in two mutually exclusive stages:

1. an independent positivity-bounded phase ensemble for each flavor and operator;
2. a screened one-gluon eikonal model with rank $n = 1, 2$:

$$\mathcal{I}_n(k_T) = \frac{C_F \alpha_s}{2\pi} \int_0^{q_{\max}} q dq \frac{[\Lambda^2/(\Lambda^2 + q^2)]^2}{q^2 + \mu_g^2} \left(\frac{q}{M_D} \right)^n \left\langle \cos(n\phi) e^{-|\mathbf{k}_T - \mathbf{q}|^2/(2w)} \right\rangle_\phi. \quad (28)$$

The rank-one term represents imaginary S-P interference; the rank-two term includes imaginary S-D and P-even/P-odd interference. Disabling the imaginary amplitude makes both functions vanish. The larger direct-phase envelope is not used to inflate the screened calculation.

7 Nucleon-level gluon dynamics

7.1 Unpolarized, helicity, and linearly polarized anchors

The canonical f_1^g central boundary uses the matched BSV19/NNPDF31 construction [20]. A separate CT18 29-pair Hessian response is propagated through AV18 smearing at five x nodes and 61 transverse-momentum points, then anchored as an uncertainty axis around the common central scheme. This preserves the distinction between the central TMD construction and collinear-PDF uncertainty.

Gluon g_1^g uses all 600 BDSSV24-NLO replicas [16]. Width variations are stored separately: they can strongly reshape the value near $k_T = 0$ while leaving the finite-grid collinear integral nearly invariant.

The T-even $h_1^{\perp g}$, g_{1T}^g , h_1^g , and $h_{1T}^{\perp g}$ arise from the retained gluon light-front overlap/OAM/Wilson parent. They are not quark TMDs with changed labels. The gluon polarization tensor and the full joint density determine the allowed interference and rank.

7.2 Naive-T-odd gluons

An early spectator-inspired model multiplied the assembled deuteron f_1^g by literature-motivated coefficients. It reproduced qualitative hierarchies but bypassed the nucleon parent, and is benchmark-only.

The canonical construction instead introduces the absorptive Wilson amplitude at the nucleon-correlator level, retains the two color classes in equation (20), and then performs the same nuclear helicity convolution as the T-even sector. External spectator and CGI-GPM calculations constrain sign, node, and hierarchy scenarios, but no released global replica set determines the absolute scale of all gluon T-odd functions. Process predictions must therefore expose C_f, C_d in equation (21).

7.3 Tensor-polarized gluons

The LL, LT, and TT gluon projections use the same shared light-front/OAM/Wilson parent, projected with the spin-1 target basis. The previously missing g_{1LT}^g and g_{1TT}^g are generated by S–D/OAM interference and screened eikonal phases, with pure-S, zero-phase, rank-covariance, link-reversal, and positivity tests. Their absolute magnitudes are predictions with named model axes.

8 Light-front deuteron and nuclear composition

8.1 Two-nucleon light-front state

The nucleonic component is

$$|D, \Lambda\rangle_{NN} = \sum_{\lambda_1, \lambda_2} \int \frac{dy d^2 \mathbf{p}_T}{2(2\pi)^3 y(1-y)} \Psi_{\Lambda; \lambda_1 \lambda_2}^{D, \text{LF}}(y, \mathbf{p}_T) |N_1(y, \mathbf{p}_T, \lambda_1) N_2(1-y, -\mathbf{p}_T, \lambda_2)\rangle. \quad (29)$$

The instant-form radial input contains

$$\Psi_{\Lambda}^{\text{IF}}(\mathbf{k}) = u_0(k) \mathcal{Y}_{00}^{1\Lambda}(\hat{\mathbf{k}}) u_2(k) \mathcal{Y}_2^{1\Lambda}(\hat{\mathbf{k}}), \quad (30)$$

followed by the declared light-front variable map and constituent Melosh rotations. The production ensemble contains AV18 (central) [11], CD-Bonn [12], and Norfolk NV-Ia, NV-Ib, NV-IIa, and NV-IIb.

The active-nucleon density is

$$\rho_{\Lambda' \Lambda; \lambda' \lambda}^{N/D}(y, \mathbf{p}_T) = \sum_{\lambda_s} \Psi_{\Lambda'; \lambda' \lambda_s}^{D, \text{LF}*}(y, \mathbf{p}_T) \Psi_{\Lambda; \lambda \lambda_s}^{D, \text{LF}}(y, \mathbf{p}_T), \quad (31)$$

with separate SS, SD, DS, and DD terms. The D state is therefore not represented only by one probability. Signed S–D interference enters tensor polarization and spin–orbit observables directly.

8.2 GTMD and TMD convolution

At zero skewness, the one-body nuclear GTMD has the schematic retained-spin form

$$W_{a/D, \Lambda' \Lambda}^{[\Gamma]}(x, \mathbf{k}_T, \Delta_T) = \sum_{N=p,n} \int \frac{dy d^2 \mathbf{p}_T}{y (2\pi)^2} \sum_{\lambda', \lambda} \rho_{\Lambda' \Lambda; \lambda' \lambda}^{N/D}(y, \mathbf{p}_T, \Delta_T) \times W_{a/N, \lambda' \lambda}^{[\Gamma]} \left(\frac{x}{y}, \mathbf{k}_T - \frac{x}{y} \mathbf{p}_T, \Delta_{T,N} \right) J(y, \mathbf{p}_T, \Delta_T). \quad (32)$$

Its forward limit gives

$$F_{a/D,I}^{\text{IA}}(x, \mathbf{k}_T) = \sum_{N=p,n} \int_x^1 \frac{dy}{y} \int \frac{d^2 \mathbf{p}_T}{(2\pi)^2} \rho_{I;\lambda'\lambda}^{N/D}(y, \mathbf{p}_T) F_{a/N,\lambda'\lambda} \left(\frac{x}{y}, \mathbf{k}_T - \frac{x}{y} \mathbf{p}_T \right) J_T. \quad (33)$$

In b_T space the transverse shift becomes a phase:

$$\tilde{F}_{a/D,I}^{\text{IA}}(x, \mathbf{b}_T) = \sum_N \int_x^1 \frac{dy}{y} \int \frac{d^2 \mathbf{p}_T}{(2\pi)^2} \rho_{I;\lambda'\lambda}^{N/D}(y, \mathbf{p}_T) e^{i(x/y) \mathbf{b}_T \cdot \mathbf{p}_T} \tilde{F}_{a/N,\lambda'\lambda}(x/y, \mathbf{b}_T) J_T. \quad (34)$$

This is the form to be consumed by complete evolution.

8.3 Binding, off-shell response, and completely positive maps

Binding and Fermi motion enter through $y, \mathbf{p}_T$ in equation (33). Off-shell, shadowing, and antishadowing responses act on the full joint spin density. The accepted implementation uses ordered completely positive maps

$$\mathcal{E}_m[\rho] = \sum_{\alpha} K_{m\alpha} \rho K_{m\alpha}^{\dagger}, \quad \sum_{\alpha} K_{m\alpha}^{\dagger} K_{m\alpha} \preceq \mathbf{1}, \quad (35)$$

so Hermiticity and positivity are preserved. The ordering is explicit because polarized response maps need not commute. These maps replace, not add to, older scalar coefficient responses.

8.4 Shadowing and antishadowing

The unpolarized small- x baseline follows the H1 diffractive PDF [23] and the Frankfurt–Guzey–Strikman double-scattering construction [22]. Schematically,

$$\delta f_{j/D}^{\text{shad}}(x, Q) \sim - \int_x^{x_{\mathbb{P},\max}} dx_{\mathbb{P}} \int dt f_{\mathbb{P}/p}(x_{\mathbb{P}}, t) f_{j/\mathbb{P}}(x/x_{\mathbb{P}}, Q) F_D(4[q_T^2 + x_{\mathbb{P}}^2 m_N^2]). \quad (36)$$

Polarized and tensor shadowing responses are independent model ratios: unpolarized diffraction does not determine them. Antishadowing is a separate mechanism constrained by the moment ledger, not an arbitrary enhancement attached to every TMD.

8.5 Pion exchange and the $NN\pi$ sector

The deuteron Fock expansion contains

$$|D\rangle = \sqrt{Z_{NN}} |NN\rangle + |NN\pi\rangle + \dots \quad (37)$$

The sourced Sullivan mechanism [24] has the longitudinal convolution

$$F_{a/D}^{(\pi)}(x, \mathbf{k}_T) = \int_x^1 \frac{dy}{y} \int d^2 \mathbf{q}_T f_{\pi/D}(y, \mathbf{q}_T; \Lambda', \Lambda) F_{a/\pi} \left(\frac{x}{y}, \mathbf{k}_T - \frac{x}{y} \mathbf{q}_T \right). \quad (38)$$

The implementation uses the Miller spin-resolved splitting information and a Vpion/JAM pion input where the corresponding channel is supported. The pion term is included once. A recent LF Hamiltonian EFT calculation does not publish the spin-resolved three-body deuteron amplitude required to replace this boundary, so no arbitrary transfer Gaussian is invented.

8.6 Other non-nucleonic sectors

Delta–Delta, SRC/cluster, and hidden-color sectors have explicit interfaces and correlated zero-centered sensitivity members. A preferred nonzero central addition requires a sourced flavor-, color-, helicity-, and transverse-resolved correlator. The present hidden-color central term is therefore excluded rather than assigned an arbitrary probability.

9 b_1 , tensor normalization, and physical reductions

The inclusive tensor distribution is

$$\delta_T q(x) = q^0(x) - \frac{1}{2}[q^{+1}(x) + q^{-1}(x)], \quad (39)$$

and at leading order

$$b_1(x, Q^2) = \frac{1}{2} \sum_q e_q^2 [\delta_T q(x, Q^2) + \delta_T \bar{q}(x, Q^2)]. \quad (40)$$

The project convention maps the named LL distribution by

$$f_{1LL} = -\frac{2}{3} \delta_T f_1, \quad (41)$$

with the sign adapter tested explicitly. The HERMES comparison [2] combines the one-body tensor impulse term with the allowed pion contribution. It is a validation of the collinear LL marginal and does not determine all tensor TMDs.

For rank-zero channels,

$$\int d^2 \mathbf{k}_T F_{a/D}^{(0)}(x, \mathbf{k}_T; Q) = f_{a/D}(x; Q). \quad (42)$$

For $r > 0$, the unweighted angular integral vanishes, while a weighted moment may remain:

$$F^{(n)}(x) = \int d^2 \mathbf{k}_T \left(\frac{k_T^2}{2M_D^2} \right)^n F(x, k_T^2). \quad (43)$$

10 Composition graph and uncertainty semantics

The total parent is not a sum of every available table. Each component is typed as

$$\mathcal{W}_{\text{canonical}} = \mathcal{W}_{\text{baseline}} \sum_{m \in \mathcal{A}} \mathcal{W}_m^{\text{additive}}, \quad \theta_e \in \mathcal{E}_e \quad \text{for exclusive alternatives and ensembles.} \quad (44)$$

The graph rejects two paths that represent the same phase, OAM interference, or nuclear response. In particular:

- the CP-map nuclear responses replace legacy coefficient responses;
- direct axial phases and screened one-gluon phases are alternatives;
- spectator-inspired downstream gluon rescaling is benchmark-only;
- the $NN\pi$ contribution is added once;
- unsupported cluster/hidden-color amplitudes are zero-centered alternatives.

Table 3: Meaning of uncertainty and sensitivity axes.

Axis	Interpretation
PDF/TMD replica	Member variation supplied by an external statistical or posterior ensemble: BDSSV24 (600), JAMDiFF (968), BPV20 (500).
Hessian	Correlated eigenvector propagation: CT18 (29 pairs) and MSHT20QED neutron CSB (38 pairs).
Published interval hull	All parameter corners of reported asymmetric intervals, used for Yang g_{1T} ; not a confidence region without the source covariance.
Wave-function envelope	AV18, CD-Bonn, NV-Ia, NV-Ib, NV-IIa, and NV-IIb dependence after normalization and common convolution.
Transverse-profile envelope	Nonperturbative k_T or b_T shape variation; it is distinct from collinear normalization uncertainty.
Gauge/color scenario	Unknown absorptive phase and f/d gluon color-link composition.
Nuclear mechanism	Off-shell, shadowing, antishadowing, pion, cluster, and SRC response choices, with channel and x-domain restrictions.
OAM/Fock scenario	Alternative allowed interference content in $L_z = 0, \pm 1, \pm 2$ sectors.
CSB power counting	Zero-centered 5% quark and 2% gluon amplitude envelope for TMDs lacking a specific calculation; not a fit uncertainty.
Numerical error	Quadrature, finite transform range, interpolation, and closure residuals.

No quadrature combination is advertised as a global confidence band. Production plots expose the named axes separately.

11 Evidence ledger

11.1 Quark and antiquark sectors

Table 4: Evidence status of the quark basis. Tensor entries apply to quarks and antiquarks with explicit flavor and proton/neutron parents.

Function	Central source	Interpretation and band
f_1	CT18NNLO; MSHT20QED CSB	Fit/phenomenology ; CT18 Hessian, sourced neutron CSB, wave/profile/nuclear axes.
g_1	BDSSV24-NLO	Fit/phenomenology ; all 600 replicas and spin-dependent nuclear convolution.
h_1	JAMDiFF w LQCD	Fit+lattice ; 968 replicas with composed-boundary positivity compatibility.
$f_{1T}^\perp$	BPV20 arTeMiDe	Fit/phenomenology ; 500 replicas, flavor dependence, staple reversal.

Function	Central source	Interpretation and band
g_{1T}	Yang et al. 2024	Fit/phenomenology central; 16-corner published-interval hull and sea/OAM sensitivity.
$h_{1L}^\perp$	JAMDiFF h_1 +WW	Fit+lattice+model ; correlated replicas and genuine-term scenario.
$h_1^\perp$	BPV20-linked hierarchy	Model ; independent Boer–Mulders operator scenario, not joint fit.
$h_{1T}^\perp$	positivity-bounded OAM	Model ; zero perturbative central plus signed hadronic members.
all LL/LT/TT	shared S–D/OAM/Wilson parent and nuclear response	Wave-function+model+phenomenology ; common-parent prediction, six-wave and phase/nuclear ensembles, b_1 where applicable.

11.2 Gluon sectors

Table 5: Evidence status of the gluon basis.

Function	Central source	Interpretation and band
f_1^g	BSV19/NNPDF31; CT18 response	Fit/phenomenology central with CT18 Hessian, wave, profile, and nuclear axes.
g_1^g	BDSSV24-NLO	Fit/phenomenology ; all 600 replicas and separate transverse-profile envelope.
all other U/L/T	gluon LF overlap/OAM/Wilson parent	Model constrained by helicity algebra, rank, f/d separation, wave functions, positivity, and literature benchmarks.
all LL/LT/TT	same shared parent with spin-1 density and nuclear maps	Model prediction with pure-S, zero-phase, link, rank, wave, and nuclear tests.

This is the intended meaning of “bringing every TMD to the f_1 level”: comparable provenance and validation discipline, not equal data precision.

12 Validation and numerical acceptance

12.1 Exact and controlled-limit tests

The implemented validation matrix includes:

1. correlator hermiticity, light-front parity, and gauge-link reversal;
2. reconstruction of all allowed target and parton polarization blocks;
3. transverse-rank covariance and physical origin limits;

4. proton/neutron traceability and the exact charge-symmetry limit;
5. pure-S, zero-phase, weak-phase, and real one-body limits;
6. independent gluon f/d components and refusal of implicit hard-color weights;
7. GTMD-to-TMD, TMD-to-PDF, GPD/form-factor, and b_1 marginals;
8. wave-function normalization, number, plus-momentum, and tensor ledgers;
9. no-double-counting closure for nuclear and non-nucleonic mechanisms.

12.2 Density positivity

For every retained joint density,

$$\rho = \rho^\dagger, \quad v^\dagger \rho v \geq 0 \quad \forall v, \quad (45)$$

is tested numerically. Positivity is imposed on the full matrix, not by clipping individual TMDs. At the WP12 scientific inspection,

$$\lambda_{\min}^{qD} = 4.12612 \times 10^{-4}, \quad \lambda_{\min}^{gD} = 2.0881 \times 10^{-2}. \quad (46)$$

No final positivity contraction was required; the common completion scale was one.

12.3 Closure and response diagnostics

The resolved parent contains 38,880 quark rows and 49,680 gluon rows. The maximum closure residual is

$$\epsilon_{\text{close}}^q = 0, \quad \epsilon_{\text{close}}^g = 1.7347 \times 10^{-18}. \quad (47)$$

The maximum CP-map recomposition shifts are approximately 2.73% and 2.91% of the local f_1 reference for quarks and gluons. The recorded future/past residual is zero for quarks and of order 10^{-9} GeV^{-2} for gluons.

12.4 Physical benchmarks

Benchmarks include CT18 and BDSSV24 marginal reproduction, JAMDiFF transversity and BPV20 Siverson member checks, the HERMES b_1 comparison, deuteron body form factors, LF current/angular-condition diagnostics, six-wave-function comparisons, rank-aware Fourier round trips, and literature node/hierarchy comparisons for model-dependent gluon functions.

The WP12-E acceptance report records 36/36 evidence rows and six final gates passing. The full repository regression after the documentation integration is 482 passing tests.

13 Scale scheme and the boundary before evolution

The in-house boundary uses a Collins square-root-soft subtraction [25, 26], the delta rapidity regulator, $\overline{\text{MS}}$ UV renormalization, and the canonical line

$$\zeta_i = \mu_i^2, \quad \mu_f = Q, \quad \zeta_f = Q^2. \quad (48)$$

The TMD equations are

$$\frac{d}{d \ln \mu} \ln \tilde{F}_a(x, b_T; \mu, \zeta) = \gamma_F^a \left(\alpha_s, \frac{\zeta}{\mu^2} \right), \quad (49)$$

$$\frac{d}{d \ln \sqrt{\zeta}} \ln \tilde{F}_a(x, b_T; \mu, \zeta) = -\mathcal{D}_a(b_T; \mu), \quad (50)$$

$$\frac{d}{d \ln \mu} \mathcal{D}_a(b_T; \mu) = \Gamma_{\text{cusp}}^a(\alpha_s). \quad (51)$$

At leading power, the Collins–Soper kernel depends on the quark or gluon color representation, not on target polarization. Polarization resides in the boundary and matching coefficients.

The present code has heterogeneous evolution depth: BPV20 uses its native optimal-TMD evolution; some rank-zero quark and gluon channels have the typed in-house CSS route; several higher-rank and tensor boundaries remain at $Q = 5$ GeV with declared profile scenarios. Putting them on one fixed- Q plot does not imply one complete evolution solution.

The next phase must evolve

{proton-in-D, neutron-in-D, sum, difference, nuclear correction, total}

for every flavor, target channel, transverse rank, link/color class, mechanism, and correlated ensemble member. It must use the appropriate J_r transforms, preserve closure after evolution, and refuse scheme mismatches.

At the observable level, a complete phase-space prediction also requires

$$\frac{d\sigma}{dP} = W(q_T, Q) + Y(q_T, Q), \quad Y = \sigma_{\text{FO}} - [W]_{\text{expanded to FO}}, \quad (52)$$

so the nonperturbative TMD boundary is not forced to mimic the high- q_T fixed-order tail. A general process-specific Y term remains open.

14 Implemented software map and reproducibility

Table 6: Principal computational objects.

Layer	Representative implementation
Spin and transverse representation	<code>spin.py</code> , <code>transverse_tensors.py</code> , <code>quark_correlator.py</code> , <code>gluon_correlator.py</code> , <code>registry.py</code> .
Nucleon quark inputs	<code>nucleon_inputs.py</code> , <code>transversity.py</code> , <code>bpv20_sivers.py</code> , <code>worm_gear_inputs.py</code> , <code>boer_mulders.py</code> .
Nucleon gluon and phases	<code>nucleon_gluon_inputs.py</code> , <code>gluon_lfwf_todd.py</code> , <code>gluon_todd.py</code> , <code>gauge_link_phase.py</code> .
Deuteron wave functions	<code>wavefunctions/av18.py</code> , <code>cd_bonn.py</code> , <code>norfolk.py</code> , <code>light_front.py</code> .
Nuclear mechanisms	<code>gtmd_convolution.py</code> , <code>operator_nuclear_response.py</code> , <code>diffractive_shadowing.py</code> , <code>pion_exchange.py</code> , <code>nonnucleonic_transverse.py</code> .

Layer	Representative implementation
Composition and resolved output	<code>canonical_composition.py</code> , <code>canonical_parent_enrichment.py</code> , <code>resolved_nuclear_parent.py</code> .
Scheme and transforms	<code>tmd_scheme.py</code> , <code>canonical_scheme.py</code> , <code>quark_tmd_matching.py</code> , <code>gluon_tmd_matching.py</code> , <code>tmd_evolution.py</code> , <code>fourier.py</code> .
Validation	<code>joint_positivity.py</code> , <code>moment_ledger.py</code> , <code>controlled_limits.py</code> , and the generated WP11/WP12 reports.

The authoritative numerical parents are

- `outputs/parent_tmds/wp12_resolved_quark_parent.csv`;
- `outputs/parent_tmds/wp12_resolved_gluon_parent.csv`;
- their `.correlators.csv` companions;
- `outputs/parent_tmds/wp12_canonical_composed_quark.csv`;
- `outputs/parent_tmds/wp12_canonical_composed_gluon.csv`.

The acceptance authority is `outputs/validation/wp12e_acceptance.json` together with the resolved-parent and evidence-parity reports.

15 Requirements for a genuinely predictive next-level model

15.1 Change of model class

The present construction is a symmetry-complete and nuclear-dressed phenomenological synthesis. Its common correlator architecture enforces consistency, but many of the scalar amplitudes entering that correlator come from different fits or from separately normalized model sectors. This is not equivalent to deriving the full correlator from one microscopic quantum state.

A genuinely predictive next-level model requires a change in the direction of inference. The present construction follows

$$\{\text{PDF/TMD fits, nuclear models, phase models}\} \longrightarrow \Phi_{a/D}^{\text{assembled}} \longrightarrow \{F_A\}. \quad (53)$$

The next-level construction must instead follow

$$\{H_{\text{LF}}^{\text{ren}}, \boldsymbol{\theta}_{\text{micro}}\} \longrightarrow \{|\Psi_N\rangle, |\Psi_D\rangle, \mathcal{U}_\eta\} \longrightarrow W_{a/N}, W_{a/D} \longrightarrow \{F_A, \mathcal{O}_{\text{process}}\}, \quad (54)$$

where one renormalized microscopic model and a limited shared parameter vector $\boldsymbol{\theta}_{\text{micro}}$ generate all flavors, polarizations, OAM interferences, gauge-link phases, nuclear components, and observables. External PDFs, TMDs, lattice matrix elements, form factors, and b_1 then calibrate or test the same parameter vector; they no longer supply unrelated named TMDs.

Even this stronger construction would be a controlled effective-QCD prediction, not an exact solution of continuum QCD. Its predictive content would come from using substantially fewer shared microscopic degrees of freedom than the number of observables predicted, and from successfully predicting quantities that were not used for calibration.

15.2 Microscopic light-front Hamiltonian and regulated Hilbert space

The first required object is a gauge-consistent, renormalized light-front Hamiltonian eigenproblem,

$$H_{\text{LF}}^{\text{ren}}(\Lambda_{\text{UV}}, \Lambda_{\text{IR}}; \boldsymbol{\theta}_{\text{QCD}}, \boldsymbol{\theta}_{\text{EFT}}) |\Psi_h(P^+, \mathbf{P}_T, \Lambda)\rangle = M_h^2 |\Psi_h(P^+, \mathbf{P}_T, \Lambda)\rangle. \quad (55)$$

The regulator, counterterms, basis truncation, and renormalization conditions must be part of the model definition. A numerical diagonalization without a cutoff study or sector-dependent renormalization is not a predictive Hamiltonian model.

For the nucleon, the minimum useful Fock expansion is

$$|N, \lambda\rangle = \psi_{qqq}^\lambda |qqq\rangle + \psi_{qqqg}^\lambda |qqqg\rangle + \psi_{qqqq\bar{q}}^\lambda |qqqq\bar{q}\rangle + \cdots, \quad (56)$$

with explicit $u, d, \bar{u}, \bar{d}$ and gluon degrees of freedom, color singlet constraints, parton helicities, and orbital sectors sufficient to support the transverse ranks in tables 1 and 2. At minimum, the basis must contain the $L_z = 0, \pm 1, \pm 2$ interferences needed by Sivers, Boer–Mulders, worm-gear, pretzelosity, linear-gluon, and spin-1 tensor projections. A valence-only $|qqq\rangle$ state cannot predict antiquark TMDs, genuine quark–gluon correlations, or the gluon sector.

The Hamiltonian or effective interaction must specify:

1. quark masses and kinetic terms in a declared scheme;
2. confining interactions and their rotational-symmetry restoration;
3. quark–gluon vertices and instantaneous light-front interactions;
4. chiral dynamics capable of producing the pion cloud and sea-flavor asymmetry;
5. counterterms and sector-dependent renormalization conditions;
6. zero-mode treatment and the consequences of Fock-space truncation;
7. a convergence sequence in longitudinal resolution, transverse basis, Fock sectors, and ultraviolet cutoff.

Candidate realizations include basis light-front quantization, a Hamiltonian effective-field-theory construction, or a continuum Bethe–Salpeter/Dyson–Schwinger amplitude mapped consistently to the light front. The project should select one primary generative route. Combining wave functions from mutually unrelated microscopic models would reproduce the present patchwork at a deeper level.

15.3 Nucleon GTMDs from common wave-function overlaps

For T-even distributions, the renormalized nucleon GTMD must be calculated as an overlap of the same Fock amplitudes. Schematically,

$$W_{a/N, \lambda' \lambda}^{[\Gamma]}(x, \mathbf{k}_T, \Delta_T) = \sum_n \sum_{j \in a} \int [dx]_n [d^2 \mathbf{k}_T]_n \delta(x - x_j) \delta^{(2)}(\mathbf{k}_T - \mathbf{k}_{Tj}) \\ \times \psi_{n, \lambda'}^*(\{x_i, \mathbf{k}'_{Ti}, \lambda_i, c_i\}) \mathcal{P}_j^{[\Gamma]} \psi_{n, \lambda}(\{x_i, \mathbf{k}_{Ti}, \lambda_i, c_i\}), \quad (57)$$

where the initial/final transverse momenta implement one common recoil convention and $\mathcal{P}_j^{[\Gamma]}$ acts on the active parton. Quark, antiquark, and gluon correlators must be derived from this same state. Their normalizations are then linked by baryon number, momentum, helicity, and angular-momentum sum rules rather than independently chosen.

The overlap implementation must demonstrate that

$$\left. \begin{array}{l} W(x, \mathbf{k}_T, \mathbf{\Delta}_T) \\ \int d^2 \mathbf{k}_T W(x, \mathbf{k}_T, \mathbf{\Delta}_T) \\ W(x, \mathbf{k}_T, \mathbf{0}) \\ \int dx d^2 \mathbf{k}_T W \end{array} \right\} \longrightarrow \left\{ \begin{array}{l} \text{GTMD} \\ \text{GPD} \\ \text{TMD} \\ \text{local current or EMT form factor} \end{array} \right. \quad (58)$$

with one regulator and one state normalization. A TMD-specific Gaussian may be used only as a quantified numerical regulator or controlled approximation, not as an independent physical width for every function.

The microscopic state must also reproduce OAM and spin decompositions. For example, in a declared gauge and scheme,

$$\frac{1}{2} = \frac{1}{2} \Delta \Sigma(\mu) + L_q(\mu) + \Delta G(\mu) + L_g(\mu), \quad (59)$$

and the canonical/kinetic OAM distinction must be related to the gauge-link definition used by the GTMD. The sign pattern of flavor-resolved Sivers or OAM-sensitive functions must emerge from these amplitudes, not be imposed as a plot-level coefficient.

15.4 Dynamical Wilson lines and naive-T-odd functions

Real light-front overlaps alone cannot predict naive-T-odd TMDs. The microscopic model must calculate, or systematically approximate, the process-dependent Wilson line

$$\mathcal{U}_\eta[0, z] = \mathcal{P} \exp \left[ig \int_{\mathcal{C}_\eta(0, z)} d\xi^\mu A_\mu^a(\xi) t^a \right], \quad (60)$$

using the same gluon fields and color couplings that appear in equation (55). An expansion such as

$$\mathcal{U}_\eta = 1 + ig \int_{\mathcal{C}_\eta} d\xi \cdot A (ig)^2 \int_{\mathcal{C}_\eta} \int_{\mathcal{C}_\eta} \mathcal{P}[A(\xi) A(\zeta)] + \dots \quad (61)$$

must have a declared truncation, rapidity regulator, color representation, and infrared treatment.

At minimum, the calculation must:

- generate the absorptive phase from explicit interference with intermediate states, rather than assign an arbitrary imaginary constant;
- recover exact future/past sign reversal where factorization requires it;
- retain distinct quark flavor and gluon f^{abc} - and d^{abc} -type color structures;
- reproduce a perturbative one-gluon limit and demonstrate stability when higher eikonal orders are included;
- identify processes for which the assumed staple factorization is established, conditional, or broken by Glauber/color entanglement.

The quark Sivers, Boer–Mulders, and all gluon T-odd functions would then be different projections of common rescattering amplitudes. Their relative signs and magnitudes would become correlated predictions. A universal phase multiplied onto unrelated operators is specifically disallowed.

15.5 Microscopic spin-1 nuclear state

The deuteron cannot be generated by convolving a microscopic nucleon with an unrelated phenomenological nuclear response. The nuclear state must be derived in a Hilbert space that resolves both nucleonic and non-nucleonic sectors:

$$|D, \Lambda\rangle = \Psi_{NN}^\Lambda |NN\rangle + \Psi_{NN\pi}^\Lambda |NN\pi\rangle + \Psi_{\Delta\Delta}^\Lambda |\Delta\Delta\rangle + \Psi_{6q}^\Lambda |6q\rangle + \dots \quad (62)$$

The amplitudes must share one normalization,

$$1 = Z_{NN} + Z_{NN\pi} + Z_{\Delta\Delta} + Z_{6q} + \dots, \quad 1 = \sum_\alpha \int [d\Gamma_\alpha] \sum_i x_i |\Psi_\alpha|^2, \quad (63)$$

so baryon number and plus momentum close without after-the-fact renormalization.

The NN sector must preserve the successful AV18/CD-Bonn/Norfolk long-distance S/D phenomenology or derive an equally accurate interaction, including consistent one- and two-body currents. The higher sectors must carry explicit spin, isospin, color, parton flavor, transverse momentum, and off-forward transfer. A scalar pion distribution or a six-quark probability is insufficient to predict tensor TMDs.

The deuteron GTMD should be evaluated as a matrix element in the full state,

$$W_{a/D}^{[\Gamma]} = \sum_{\alpha, \beta} \langle \Psi_D^\alpha | \hat{\mathcal{O}}_a^{[\Gamma]} [\mathcal{U}_\eta] | \Psi_D^\beta \rangle, \quad (64)$$

including diagonal contributions and physically allowed interference between Fock sectors. Impulse, exchange-current, shadowing, and non-nucleonic terminology should then label controlled reductions of equation (64), not independently normalized additions.

At small x , coherent multiple scattering must be derived from the same nucleon diffractive amplitudes and nuclear state. Polarized and tensor shadowing cannot be copied from the unpolarized response; their helicity amplitudes must be calculated. At intermediate and large x , off-shell, Fermi-motion, SRC, and EMC-like behavior must emerge from the joint spectral/partonic amplitude with a tested separation of scales.

15.6 Matching, renormalization, and evolution

A low-resolution Hamiltonian wave function is not by itself a TMD in a specified QCD scheme. A complete prediction requires a matching map

$$\widetilde{W}_{a/h}^{\text{QCD}}(x, b_T; \mu, \zeta) = \mathcal{Z}_a^{\text{LF} \rightarrow \text{TMD}}(b_T; \Lambda_{\text{LF}}, \mu, \zeta) \otimes \widetilde{W}_{a/h}^{\text{LF}}(x, b_T; \Lambda_{\text{LF}}) + \mathcal{O}\left(\frac{\Lambda_{\text{LF}}}{Q}\right), \quad (65)$$

including soft subtraction, rapidity renormalization, and operator mixing. The matching must identify which Hamiltonian regulator dependence cancels against $\mathcal{Z}^{\text{LF} \rightarrow \text{TMD}}$.

The next-level model must use one consistent route for:

1. nonsinglet, singlet, and quark–gluon collinear evolution;
2. quark and gluon Collins–Soper kernels;
3. transverse ranks $r = 0, 1, 2, \dots$;
4. T-even and T-odd operators in their correct gauge-link classes;
5. threshold matching and heavy flavors when the Q range requires them;

6. process-level $W + Y$ matching.

Fit-native evolved inputs may remain as external validation data, but they must not replace the evolution of the microscopic state.

15.7 Calibration without rebuilding a patchwork

The microscopic parameter vector must be small, shared, and identifiable. The likelihood should be defined at the observable or well-defined matrix element level:

$$p(\boldsymbol{\theta}_{\text{micro}}|\mathcal{D}) \propto p(\mathcal{D}|\boldsymbol{\theta}_{\text{micro}})p_{\text{EFT}}(\boldsymbol{\theta}_{\text{micro}})p_{\text{ren}}(\boldsymbol{\theta}_{\text{micro}}), \quad (66)$$

with correlated experimental, lattice, truncation, regulator, and numerical covariances. Suitable calibration data include:

- hadron masses, charges, magnetic moments, and axial charges;
- nucleon electromagnetic, axial, and energy–momentum form factors;
- selected unpolarized/helicity PDFs and low Mellin moments;
- lattice quasi-/pseudo-distributions and GTMD-related moments where scheme matching is controlled;
- deuteron binding, asymptotic S/D normalization, elastic form factors, quadrupole and magnetic moments;
- a restricted subset of b_1 , SIDIS, and T-odd data sufficient to constrain common phases.

The calibration must not assign an independent free normalization to each TMD. A parameter associated with a quark–gluon vertex, OAM mixing, or eikonal kernel must affect every correlator projection to which that microscopic interaction contributes. Identifiability should be assessed through posterior correlations, Fisher information or profile likelihoods, and closure on withheld observables.

The fit must be divided into calibration and prediction sets before final optimization. A quantity counts as a prediction only if its normalization and shape were not directly fitted. For example, using Sivers data to fix a common rescattering kernel and then predicting Boer–Mulders, tensor T-odd, and gluon color-sector observables is a meaningful test; fitting each of those independently is not.

15.8 Uncertainty budget for a generative model

The next-level uncertainty is not a band assembled from unrelated models. It must propagate a common ensemble of microscopic states. For any observable $\mathcal{O}$,

$$\text{Cov}(\mathcal{O}_i, \mathcal{O}_j) = \left\langle [\mathcal{O}_i(\boldsymbol{\theta}, \nu) - \overline{\mathcal{O}}_i] [\mathcal{O}_j(\boldsymbol{\theta}, \nu) - \overline{\mathcal{O}}_j] \right\rangle_{\boldsymbol{\theta}, \nu}, \quad (67)$$

where $\boldsymbol{\theta}$ denotes calibrated microscopic parameters and ν collects controlled truncation, regulator, matching, and numerical choices. The covariance must retain cross-flavor, cross-TMD, quark–gluon, proton–neutron, and cross-observable correlations.

The error budget must separately report:

1. statistical/calibration uncertainty;
2. Hamiltonian EFT truncation;
3. Fock-sector and basis truncation;

4. ultraviolet/infrared regulator and renormalization uncertainty;
5. Wilson-line/eikonal truncation;
6. perturbative matching and scale variation;
7. TMD evolution and large- b_T uncertainty;
8. nuclear many-body truncation;
9. numerical discretization and transform error.

Convergence in one category cannot be used to hide another. In particular, positivity is a constraint on allowed states, not an uncertainty estimate.

15.9 Validation ladder and falsifiability

The model should be validated in increasing dynamical depth:

1. **Algebraic layer:** color singlets, Poincare/light-front kinematics, Hermiticity, parity, time reversal with link reversal, permutation symmetry, and exact spin-1 projection closure.
2. **State layer:** normalization, regulator independence within truncation errors, mass spectrum, rotational multiplets, and convergence of Fock probabilities and OAM components.
3. **Local-current layer:** charges, magnetic moments, axial charges, energy-momentum moments, and Ward identities using currents consistent with the Hamiltonian.
4. **Marginal layer:** simultaneous TMD/GPD/PDF/form-factor reductions from the same GTMD.
5. **Nucleon layer:** withheld flavor-resolved PDFs, TMD moments, lattice observables, and process sign changes.
6. **Deuteron layer:** binding, elastic observables, b_1 , tagged structure, tensor moments, and controlled proton/neutron limits.
7. **Process layer:** SIDIS, DY, and gluon-sensitive predictions with explicit hard factors, fragmentation, color weights, and $W + Y$.

Failure of a withheld observable must update or reject the microscopic model. It must not be repaired by adding a TMD-specific coefficient unless that coefficient corresponds to a missing operator added consistently to the Hamiltonian or current.

15.10 Required software architecture and deliverables

The present modular project can host the new model, but the central state must change. Required new computational objects are:

1. a versioned **MicroscopicHamiltonian** carrying regulator, counterterm, basis, and renormalization metadata;
2. a sparse/distributed **FockState** representation with parton flavor, color, helicity, momentum, OAM, and sector identity;
3. eigensolvers and convergence manifests for nucleon and deuteron states;
4. common quark/gluon GTMD overlap evaluators at nonzero Δ_T ;
5. a Wilson-line engine with perturbative/eikonal order and link/color labels;

6. a matching adapter from the Hamiltonian regulator to the declared soft-subtracted TMD scheme;
7. an ensemble store that preserves microscopic member identity through every observable;
8. automatic-differentiation or adjoint sensitivities for calibration;
9. observable-level likelihoods and withheld-validation reports.

PennyLane or another quantum simulator would be justified only if it represents a defined truncated Fock Hilbert space, implements the same Hamiltonian matrix elements, and is benchmarked against exact diagonalization on smaller sectors. A quantum circuit that merely reproduces Clebsch–Gordan coupling does not add predictive physics.

15.11 Acceptance criteria

The next-level model may be called genuinely predictive only when all of the following are satisfied:

1. One declared microscopic Hamiltonian/state ensemble generates quark, antiquark, and gluon correlators; fitted named-TMD curves are comparison data, not central inputs.
2. The nucleon state contains the Fock sectors needed for sea and gluon observables, with demonstrated basis and sector convergence.
3. Proton and neutron flavor differences, helicity, transversity, OAM, and spin–orbit correlations emerge from the state.
4. T-even GTMDs and all their TMD/GPD/PDF/form-factor reductions close from the same overlap.
5. T-odd phases are generated by an explicit Wilson-line/rescattering calculation with correct process and f/d color dependence.
6. The spin-1 state contains normalized NN and supported non-nucleonic sectors with explicit tensor polarization and interference.
7. Currents and operators are matched consistently to the Hamiltonian and satisfy Ward identities and known local limits.
8. One QCD TMD scheme and rank-aware evolution route carries the full state to multiple Q values.
9. A shared microscopic posterior predicts withheld nucleon and deuteron observables with correlated uncertainties.
10. At least one nontrivial family not used in calibration—for example tensor TMDs, Boer–Mulders from a Sivers-calibrated kernel, or gluon f/d-sensitive observables—is successfully predicted.
11. Removing any presently phenomenological TMD input does not make the central microscopic calculation undefined.
12. All symmetry, positivity, marginal, sum-rule, regulator, truncation, numerical, and no-double-counting gates pass.

This program is substantially more demanding than completing evolution of the current boundary. Evolution remains useful for the present phenomenological model, but it must not be presented as the step that turns that model into a microscopic prediction. The Hamiltonian/Fock-state and Wilson-line program above is the actual transition in model class.

Table 7: Distinction between the present synthesis and the required next-level generative model.

Question	Present model	Genuinely predictive model
Source of TMD amplitudes	Different fits and explicit model sectors	Common microscopic nucleon/deuteron state
Flavor and spin	Retained and constrained after input assembly	Generated by Hamiltonian eigenstates
T-odd phases	Fit input or screened/eikonal scenarios	Wilson-line dynamics using the model gluon field
Tensor nuclear physics	Realistic wave functions plus operator response maps	Full spin-resolved nuclear Fock-state matrix element
Non-nucleonic sectors	Sourced additions or zero-centered interfaces	Normalized amplitudes in the same eigenstate
Uncertainty	Named heterogeneous axes	Correlated microscopic posterior plus truncation errors
Evolution	Heterogeneous fixed- Q boundary routes	One matched, rank-aware QCD scheme
Meaning of prediction	Constrained extrapolation or model scenario	Withheld observable from shared calibrated parameters

16 Completion audit and remaining physics

16.1 Completed at the declared boundary

- All 18 quark/antiquark and 18 gluon forward projections.
- Explicit $u, d, \bar{u}, \bar{d}$, proton, and neutron parents.
- U/L/T/LL/LT/TT target structure and retained parton helicity.
- Independent gluon f/d gauge-link/color components.
- A shared GTMD/correlator parent and tested projections.
- Six realistic deuteron wave functions with S/D and SS/SD/DS/DD interference.
- Spin, OAM, spin-orbit, and absorptive Wilson-line mechanisms.
- Binding/Fermi motion, off-shell, shadowing, antishadowing, pion, and controlled non-nucleonic interfaces.
- Named fit, Hessian, interval, wave, profile, phase, nuclear, OAM, CSB, and numerical axes.
- Symmetry, positivity, support, marginal, moment, closure, controlled limit, and benchmark validation.

16.2 Model-dominated but honestly represented

Boer–Mulders normalization, pretzelosity, genuine WW breaking, most tensor-polarized TMDs, polarized/tensor shadowing, absolute non-Sivers gluon T-odd magnitudes, process-specific f/d mixtures, and intrinsic non-nucleonic transverse/color structure remain model dominated. They are not omitted, but neither are their bands presented as fitted confidence regions.

16.3 Open requirements after this boundary

1. Complete rank-aware, common-scheme, multi- Q evolution with resolved-component and correlated-ensemble closure.
2. Process-specific $W + Y$ matching and hard/color factors.
3. A global inference program combining tensor DIS, SIDIS, DY, lattice constraints, and gluon-sensitive observables.
4. Full nonzero-skewness GTMD phenomenology and polynomiality tests.
5. Improved polarized/tensor diffractive inputs and a spin-resolved transverse $NN\pi$ amplitude.
6. Direct fit or lattice information for tensor and gluon TMDs that currently rely on shared-parent model predictions.

17 Conclusion

The original GTMD-first proposal supplied the right organizing principle, but the route to an adequate numerical boundary required two substantive corrections. Exact zeros in a real scalar one-body parent were recognized as missing-amplitude limits, and a later smooth reduced-amplitude atlas was recognized as a bookkeeping closure rather than a derivation from the light-front parent. The present model restores the parent chain, preserves flavor and constituent identity, represents vector and tensor spin-1 structure, introduces physically distinct OAM and gauge-link amplitudes, and composes nuclear and non-nucleonic mechanisms under an explicit no-double-counting contract.

The strongest sectors inherit modern unpolarized, helicity, transversity, and Sivers ensembles. Less constrained tensor and gluon sectors are predictions of the same spin/OAM/Wilson parent with explicit sensitivity axes. The final deuteron may approach isoscalar equality, but this emerges from unequal proton and neutron flavor parents rather than being built in as a flavor-independent ansatz. All declared projections now meet the same standard of provenance, replacement interfaces, nuclear propagation, and validation, although not the same experimental precision.

This is a scientifically defensible boundary on which to build complete evolution. Evolution must transport, rather than erase, the flavor, spin, rank, color/link, constituent, mechanism, and uncertainty structure documented here.

A Machine-readable acceptance values

Quantity	Accepted value
Declared boundary	$Q = 5 \text{ GeV}$, $x_N = \{0.02, 0.05, 0.10, 0.20, 0.40\}$
Basis	18 quark/antiquark projections per flavor; 18 gluon projections
Resolved parent size	38,880 quark rows; 49,680 gluon rows
Evidence parity	36/36 rows pass
Minimum density eigenvalues	4.12612×10^{-4} (quark), 2.0881×10^{-2} (gluon)
Maximum parent closure residual	0 (quark), 1.7347×10^{-18} (gluon)
Maximum CP recomposition shift	2.73% (quark), 2.91% (gluon), relative to local f_1
Full regression	482 tests pass

B Reproduction commands

```
# Compile this manuscript.
TECTONIC_CACHE_DIR=/private/tmp/deuteron-tectonic-cache \
/Users/dustin/work/DeuteronWigner/.conda-latex/bin/tectonic \
-X compile references/model_construction_note.tex \
--outdir output/pdf
```

```
# Run the complete repository validation.
PYTHONPATH=src MPLCONFIGDIR=/private/tmp/deuteron-mpl \
/Users/dustin/miniforge3/bin/python3.9 -m pytest -q
```

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